HCC1395: somatic mutation and RNA evidence

Whole source-call audit | matched-normal DNA baseline | original tumor RNA | pinned reference ORFs. Counts below describe the specified public call sets and RNA run, not every biological mutation or a validated full-length isoform.

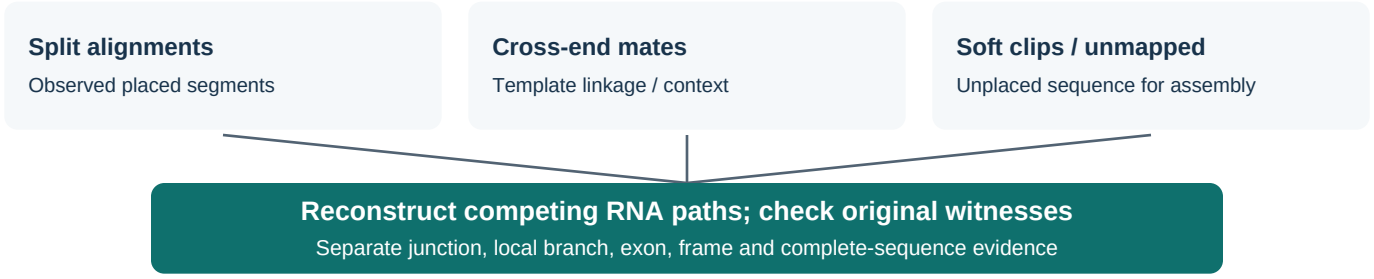
Small variants: DNA to RNA and local protein



Boxes show subset order; widths are schematic, not scaled to count.

Structural variants: complementary routes to RNA reconstruction

1,788 nominated SV records; 555 have RNA near an end



The legacy split/mate screen is one route, not an eligibility gate.

Soft-clip counts were not enumerated in the old atlas; this channel is not zero.

Breakend-window coverage is locus coverage. Split reads, cross-end mates and soft clips are complementary discovery evidence; none alone resolves a coding frame. Exon overlap is an annotation, not an eligibility filter. The older exon-restricted split/mate screen and selected junction replay are documented later without treating their negatives as absence. The newer COLO829 workflow uses all retained paths and exon-anchored clips.

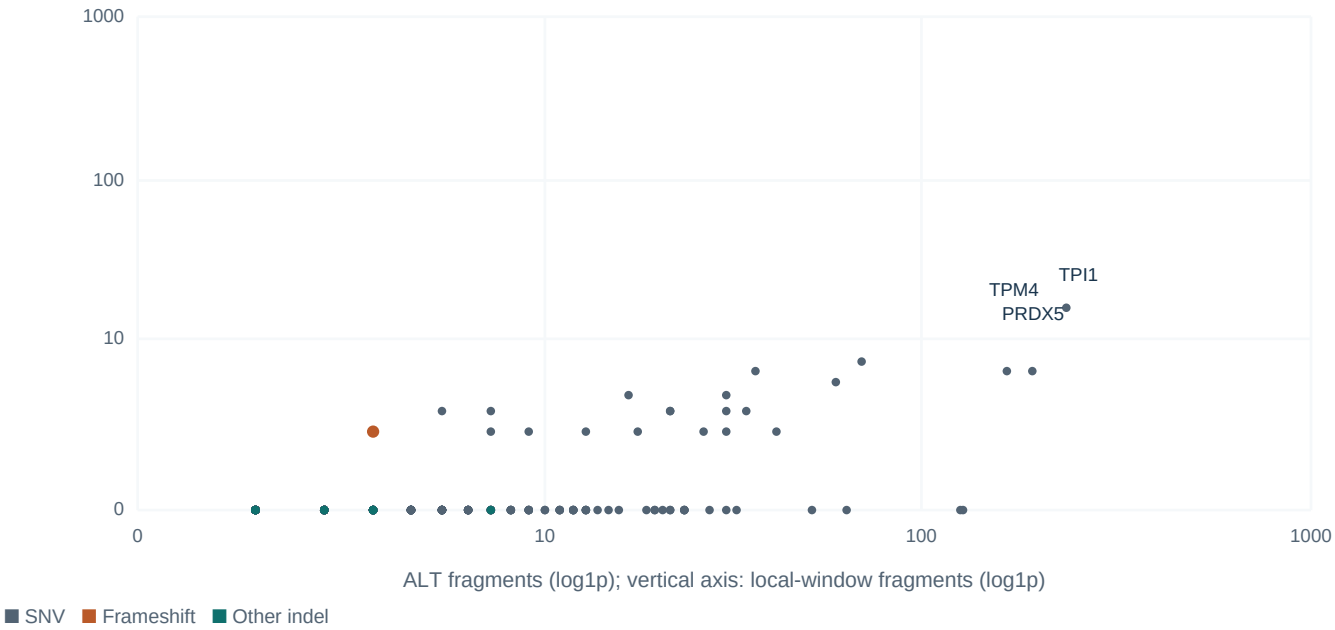
Variant landscape: depth, coding recovery and unresolved categories

Every nominated small variant is counted above. Dots show original-atlas outcomes: baseline dots had no local protein context under those settings. The labelled open square adds the new default-coverage ncORF result where present; original counts stay unchanged. A context may be reference-identical. Cultures and assays remain separate.

DNA inventory composition and original-RNA outcomes

	DNA rows	ALT calls	Atlas context
SNV	39,447	333	21
insertion	157	1	1
deletion	1,468	9	0

RNA allele support versus support for the top local protein window



Category	What this report can establish
Small variants	Original-RNA allele observations and bounded local coding windows
SVs / fusions	Original joins and competing RNA paths; annotated coding frame only where resolved
ncORFs	Validated reference ORF frames and mutation-supporting RNA; candidate translation remains conditional
Viral / other antigen sources	Not surveyed in these host-focused inputs; no absence claim

At a glance

Evidence layer	Count	Interpretation
Ingested small variants	41,072	39,447 SNVs; 1,625 indels
Ingested SV source records	1,788	DNA breakpoint records/events, counted in their source sets
SVs involving an annotated exon	354	Span for DEL/DUP/INV; nominated breakpoint or insertion flanks otherwise
Any annotated exon involved	1,728	At least one changed base or insertion flank overlaps an Ensembl 87 exon
Ensembl 87 coding effect	491	Any annotated protein-coding transcript, reference model
orfdata ncORF coding-block overlap	17	Candidate reference ORF overlap; distinct from mutant translation
Original RNA ALT-observed calls	343	2,681 ALT reads; 2,681 ALT fragments
Atlas local protein contexts	22	119 ranked contexts; new ncORF pass reported separately

Coding annotation and its denominator

Of 41,072 DNA small-variant records, 491 change the coding sequence of at least one Ensembl 87 transcript. The pinned orfdata 0.34.1 GRCh38 catalog (7,264 reference ncORF models; release 2026-10-11.1) has exact coding-block overlaps at 17 calls. The groups overlap at 3 calls, so the union is 505. These are annotation opportunities. The catalog does not assess a cell-line-specific mutant ORF, RNA expression, or peptide evidence.

Reference annotation	All DNA calls	With original RNA ALT
Ensembl coding effect	491	135
orfdata coding-block overlap	17	5
Union of either category	505	see per-call table

SNVs query their changed reference base. Deletions query removed reference bases; insertions use the VCF anchor as a candidate overlap. The per-call table preserves that rule. A coding-block hit does not prove the inserted sequence is in frame.

Any annotated exon is involved in 1,728 calls (4.2% of the small-variant source set). This includes UTR and noncoding-transcript exons. Deletions query every removed reference base, including exon crossings with intronic endpoints; insertions query both neighboring reference bases as candidate exon involvement.

Original RNA and local protein evidence

RNA assay: SRR23456709, ONT direct RNA. Each nominated small variant has a status from this run. An ALT read is a sequence observation; it is not a full-length mutant isoform.

RNA allele status	Small-variant records
alternate observed	343
reference only	239
reference and other only	53
other allele only	19
alignment without allele call	7,300
no alignment overlap	33,118

The 343 ALT-observed calls contribute 2,681 allele-read and 2,681 fragment observations. Isovar resolves at least one read-compatible **local** protein context for 22 calls (119 ranked rows). The remaining 321 ALT-observed calls have no local translation under the stated method. No-overlap and uncallable statuses cannot establish absence of RNA expression.

OSTC F9L and PRDX5 F157L have original-read local contexts. An independent proteomic study observed the mutant residues, not these entire contexts. The bounded ONT cDNA fixture is excluded from whole-run counts.

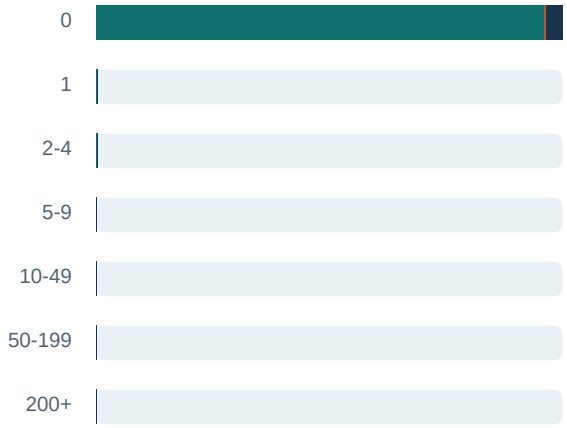
Structural-variant RNA screen

Breakend screen outcome	Source records
No breakend-window alignment	1,233
One end covered; no split candidate	149
Both ends covered; no split candidate	400
Split candidate; path unresolved	6

Six Talsania records have same-read split-alignment candidates (17 record-level observations). Published POU2F1-NME7 has one candidate read. Junction bases, orientation, splice choice and ORF remain unresolved; no fusion protein is called.

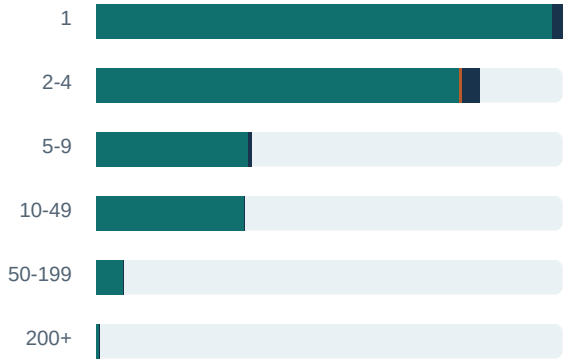
RNA alternate-read support by variant class

ALT reads per variant: all records



	SNV	insertion	deletion
0	39,114	156	1,459
1	132	0	3
2-4	105	1	5
5-9	44	0	1
10-49	43	0	0
50-199	8	0	0
200+	1	0	0

ALT reads per variant: positive support only

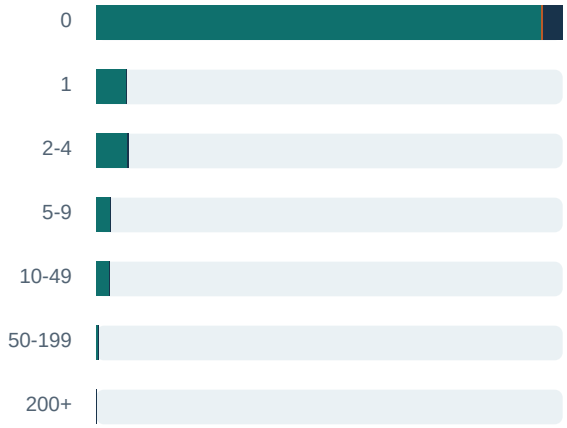


	SNV	insertion	deletion
1	132	0	3
2-4	105	1	5
5-9	44	0	1
10-49	43	0	0
50-199	8	0	0
200+	1	0	0

The positive-support panel excludes zero-ALT records. Bars stack SNV, insertion, deletion and other-replacement record counts. Matching colored columns print each class count directly. These are variant counts, not read counts.

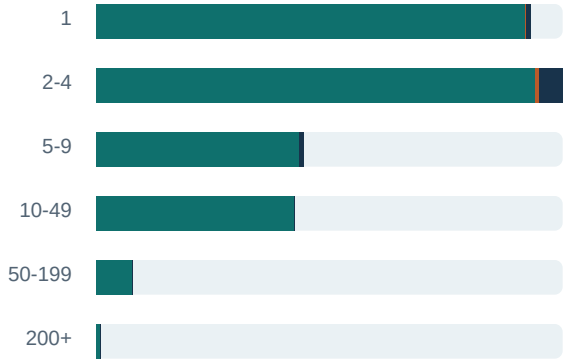
RNA ALT reads: exon-involved small variants

ALT reads per exon-involved variant: all records



	SNV	insertion	deletion
0	1,371	4	61
1	93	0	1
2-4	95	1	5
5-9	44	0	1
10-49	43	0	0
50-199	8	0	0
200+	1	0	0

ALT reads per exon-involved variant: positive support only



	SNV	insertion	deletion
1	93	0	1
2-4	95	1	5
5-9	44	0	1
10-49	43	0	0
50-199	8	0	0
200+	1	0	0

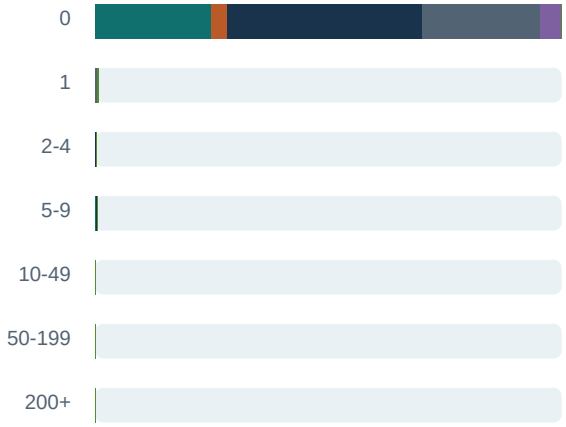
The input is the exon-involved subset of the small-variant source calls. The first panel retains zero-ALT calls; the second conditions on positive ALT reads. Insertion flanks are candidate exon involvement. SVs have a separate plot because a split alignment is not an ALT-spanning read.

Exon-involved nominated SVs: RNA breakpoint screen

354 of 1,788 nominated SV source records involve an Ensembl 87 exon. Intrachromosomal DEL/DUP/INV records query their full nominated span; INS queries insertion flanks; TRA/BND query nominated breakpoints. 0 records have insufficient breakpoint coordinates to rule out exon involvement and are excluded from this plot.

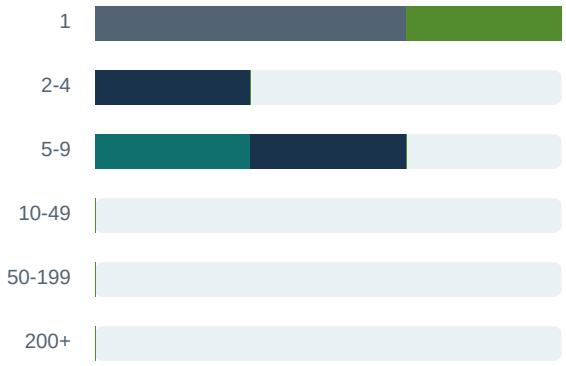
RNA screen	Exonic SV records	Candidate reads or pairs
Same-read split alignments	6	17
Cross-end primary mates	0	0

RNA link candidates per exon-involved SV: all records



	DEL	INS	DUP	INV	TRA	BND
0	87	12	145	88	16	0
1	0	0	0	2	0	1
2-4	0	0	1	0	0	0
5-9	1	0	1	0	0	0
10-49	0	0	0	0	0	0
50-199	0	0	0	0	0	0
200+	0	0	0	0	0	0

RNA link candidates per exon-involved SV: positive support only



	DEL	INS	DUP	INV	TRA	BND
1	0	0	0	2	0	1
2-4	0	0	1	0	0	0
5-9	1	0	1	0	0	0
10-49	0	0	0	0	0	0
50-199	0	0	0	0	0	0
200+	0	0	0	0	0	0

Bars count SV source records by reported class, binned on split-read plus cross-end mate candidate counts. Same-read supplementary alignments and opposite primary mates are different kinds of screening evidence; neither validates a junction sequence or translated product. The screen uses 500-base windows and mapping quality at least 20. The COLO829 benchmark call sets remain separate in the underlying table.

Original-read replay: junction versus RNA path

This selected-event replay uses oriented DNA adjacencies and actual RNA alignment records, including soft-clipped bases and low-MAPQ pieces. Direct junction fragments count reads crossing the same RNA join, even when their wider transcript sequences differ. MAPQ 20 is a sensitivity subset, not a gate on whether the junction sequence was observed. Isovar extends these joins through overlapping original read sequences; one read need not span the assembled context. Context length is a bounded assembly result, not an isoform-support count.

Nominated DNA record	Class	RNA join, all MAPQ	Join, MAPQ 20 subset	Longest context (bp)
chr13:52,688,329-60,966,095 **SUGT1** chr13:52688329-52688329 chr13:60966095-60966095	DEL	4	3	2430
chr21:42,975,947-43,035,721 **PKNOX1** chr21:42975947-42975947 chr21:43035721-43035721	DUP	2	2	2225
chr9:83,969,153-84,015,113 **HNRNPK** chr9:83969153-83969153 chr9:84015113-84015113	DUP	7	7	2198

Gene names overlap the source breakends (Ensembl 87); they do not establish a coding fusion. IGV loci are one-based inclusive, GRCh38, with no digit separators.

The chr9 duplication has seven direct junction fragments; three event paths survive assembly and the search reaches its ten-path cap. The chr13 deletion has four direct fragments, including one lost at MAPQ 20; chr21 has two. Overlapping reads extend each candidate context; their local branch support still needs evaluation. The three chr1 screen rows reuse one complex read and were not oriented here.

The HCC1395 replay BAM is a source-derived 309-record regional fixture; COLO829 uses the 129,534-record selected-site BAM. Both were extracted after aligning the complete original RNA run. No transcript models were supplied in this geometry audit, so frame and SV-derived protein status remain unresolved. Other splicing and DNA-to-RNA causation have not been excluded.

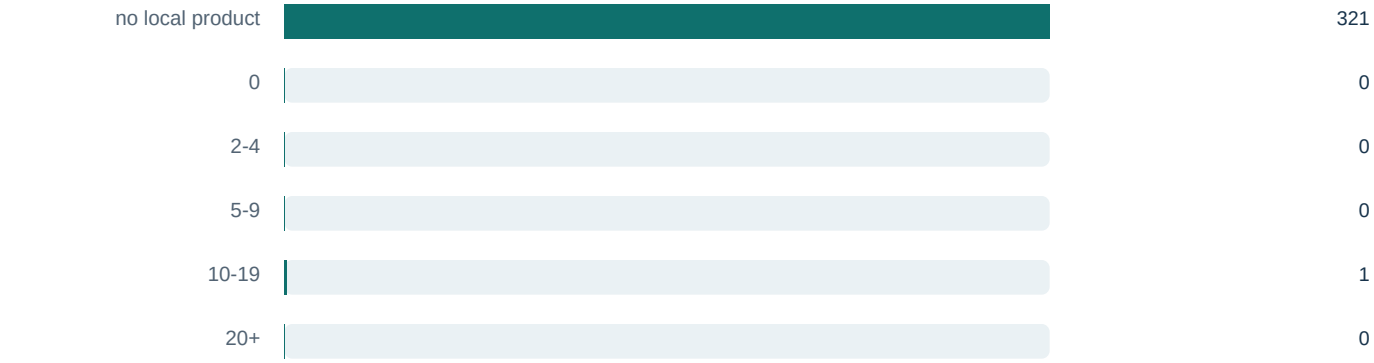
Isovar-marked amino-acid span

For each RNA ALT-observed call, count amino-acid positions in the mutation interval of its top-ranked local protein context. This is not an independently verified count of changed residues: synonymous DNA calls can produce a reference-identical amino acid window. Calls with no local context are shown explicitly. The second panel omits the one-residue category to expose the longer tail; its denominator is smaller. A zero-length interval can represent a deletion junction. These windows do not determine a complete frameshift tail or a full-length isoform.

Marked AA positions: all ALT-observed calls



Marked AA positions: excluding one-AA calls



Highest-support SNVs with local coding context

Top 3 of 21 ALT-supported calls in this category, ranked by source-library ALT fragments. Window support is reported separately. A reference-identical sequence is a coding context, not a neoORF. DNA effect labels and the displayed reference may use different transcript isoforms.

TPI1 | chr12:6867653:G>C

IGV (GRCh38): chr12:6867653-6867653 | Silent p.L66= | 236 ALT fragments

16 reported window fragments; 20 aa. Transcript(s): ENST00000229270;ENST00000396705;ENST00000613953

REF	KQSLGELIGTLNAAKVPADT
RNA	KQNLGSSSALSTRPRCRPTP

RNA differences have unclassified provenance; DNA-codon-anchored comparison only. Reference: ENST00000229270, residue 56. No unobserved sequence is filled from reference.

PRDX5 | COSM32759

IGV (GRCh38): chr11:64320895-64320895 | Substitution p.F157L | 193 ALT fragments

6 reported window fragments; 20 aa. Transcript(s): ENST00000265462

REF	VRLLADPTGAFGKETDLLLD
RNA	VRLLADPTGALGKETDLLLD

Differing focal RNA residues highlighted. Reference: ENST00000265462, residue 147. No unobserved sequence is filled from reference.

TPM4 | COSM32783

IGV (GRCh38): chr19:16093699-16093699 | Substitution p.E240Q | 166 ALT fragments

6 reported window fragments; 20 aa. Transcript(s): ENST00000300933;ENST00000344824

REF	DKLKEAETRAEFAERTVAKL
RNA	DKLKEAETRAQFAERTVAKL

Differing focal RNA residues highlighted. Reference: ENST00000300933, residue 194. No unobserved sequence is filled from reference.

Top RNA-supported frameshift nominations

Top 1 of 1 ALT-supported calls in this category, ranked by source-library ALT fragments. Window support is reported separately. A reference-identical sequence is a coding context, not a neoORF. DNA effect labels and the displayed reference may use different transcript isoforms.

C8orf76 | chr8:123220221:A>AGGGCTAC

IGV (GRCh38): chr8:123220221-123220221 | FrameShift p.L342fs | 3 ALT fragments

2 reported window fragments; 25 aa. Transcript(s): ENST00000276704

REF	**HPEVKCVGSVALTALVTVSSEEFED**
RNA	**HPEVKCVGSVARSPDCLGDCILRRI**

Differing focal RNA residues highlighted. Reference: ENST00000276704, residue 331. No unobserved sequence is filled from reference.

Top other insertions, deletions and replacements

Top 3 of 9 ALT-supported calls in this category, ranked by source-library ALT fragments. Window support is reported separately. A reference-identical sequence is a coding context, not a neoORF. DNA effect labels and the displayed reference may use different transcript isoforms.

SFXN1 | chr5:175528598:GA>G

IGV (GRCh38): chr5:175528598-175528599 | ThreePrimeUTR 3' UTR | 7 ALT fragments

RNA ALT observed, but no local protein context was reconstructed. No protein sequence is inferred.

PPIG | chr2:169641207:ATTATATTTTTTTAATGAAGTTTAGAAAAAAAGCTG>A

IGV (GRCh38): chr2:169641207-169641242 | ThreePrimeUTR 3' UTR | 3 ALT fragments

RNA ALT observed, but no local protein context was reconstructed. No protein sequence is inferred.

CNOT9 | chr2:218594451:GTCTTGAGGACCTGGGCTCCCTCTGCTA>G

IGV (GRCh38): chr2:218594451-218594478 | ThreePrimeUTR 3' UTR | 3 ALT fragments

RNA ALT observed, but no local protein context was reconstructed. No protein sequence is inferred.

SV-derived RNA transcripts: junctions, branches and unresolved frames

These are sequence-supported RNA paths. A breakpoint-associated transcript is not automatically a coding fusion or a DNA-causal product. Split placement, soft clips, original query sequence and exon anchors are complementary evidence.

HNRNPK | chr9_83969153_chr9_84015113_-+

IGV (GRCh38): chr9:83969153-83969153 chr9:84015113-84015113

7 original junction fragments; 3 pinned example RNA paths; representative local-window support 3 fragments. These measures are not pooled.

RNA junction context (nucleotides; no protein frame assigned):

`GGGCGTGCAAAACTAACATTGAAACAGTGAAAAAGAGGATTGCT`
The pinned exon-path audit retains ordinary-transcript and ambiguous-frame interpretations. No event-linked protein is established by these paths.

SUGT1 | chr13_52688329_chr13_60966095_+-

IGV (GRCh38): chr13:52688329-52688329 chr13:60966095-60966095

4 original junction fragments; 1 pinned example RNA paths; representative local-window support 1 fragments. These measures are not pooled.

RNA junction context (nucleotides; no protein frame assigned):

`CTTCTTCTTTAAGTCTTTTATCTTCCCAGAGTATTAAAGCATTC`
The pinned exon-path audit retains ordinary-transcript and ambiguous-frame interpretations. No event-linked protein is established by these paths.

PKNOX1 | chr21_42975947_chr21_43035721_-+

IGV (GRCh38): chr21:42975947-42975947 chr21:43035721-43035721

2 original junction fragments; 2 pinned example RNA paths; representative local-window support 1 fragments. These measures are not pooled.

RNA junction context (nucleotides; no protein frame assigned):

`TGGTCTCTACTAAAAATAAAAAATGTGAACCTTGCCAGTATAG`
The pinned exon-path audit retains ordinary-transcript and ambiguous-frame interpretations. No event-linked protein is established by these paths.

Coding-fusion exemplar

EIF3K-CYP39A1: separate supplied-fusion input resolves a 531-aa coding candidate with 149 Illumina junction fragments and independent ONT support. The appended deep dive shows full parental reference alignments and exact per-codon/assembly coverage. It is outside the benchmark-SV funnel.

ncORFs and other antigen-source categories

5 ALT-supported calls → 6 validated ncORF models → 0 amino-acid-changing models at default coverage. The separate single-read sensitivity pass adds 2 low-support model(s). All input models are assessed; ordinary CDS competitors remain in the exported results. Alternative ORFs on one call reuse its molecules and are not separate mutations.

Gene / IGV (GRCh38)	Reference ORF	ALT	RNA result
NELFB chr9:137255474-137255474	c9norep167	22	Reference start codon affected; no new initiation inferred
UQCC1 chr20:35384132-35384132	c20norep55	5	No retained RNA/frame match under either budget
MIR762HG chr16:30894580-30894580	c16riboseqorf60	4	Sensitivity only: 1-fragment candidate
MIR762HG chr16:30894580-30894580	c16riboseqorf61	4	RNA path incompatible with this isoform
PPIP5K2 chr5:103129447-103129447	c5norep99	3	Sensitivity only: 1-fragment candidate
ST3GAL1 chr8:133459437-133459437	c8norep145	2	No retained RNA/frame match under either budget

What prevents stronger protein calls

The 22 NELFB ALT fragments change a reference start codon. They support the RNA allele, but cannot establish an alternative initiation site. The ordinary coding-frame competitor remains in the export. For UQCC1 and ST3GAL1, relaxing coverage alone does not produce a retained RNA/frame match. MIR762HG supports one catalogue isoform path; the other path is incompatible with these reads.

The two sensitivity examples on the next page each depend on one original ONT molecule. Their nonfocal differences can include sequencing errors. They identify follow-up targets, not high-confidence new proteins. The recovered input is ALT-witness selected, so it supplies no reference-depth or allele-frequency denominator.

Sources: ORFdata 7,264-model catalogue 2026-10-11.1; complete ORF paths/proteins validated against Ensembl 87. COLO829 uses the whole-run selected-site BAM. HCC1395 uses all 36 original ALT witnesses for these loci, recovered from SRA and realigned genome-wide with minimap2 2.26; all five ALT counts reproduce. Default minimum read coverage is 2; the sensitivity pass uses 1 and reuses the same molecules. Full sequences, model alternatives, original witnesses, source hashes and budgets: ncORF audit manifest.

Assessment limits: RNA and reference Ribo-seq/MS annotations do not measure mutant translation or HLA presentation. General cryptic initiation and retained-intron ORFs remain unresolved beyond the focused DDX17 audit. Viral RNA/ORFs and cancer/testis isoform surveys are not assessed. Germline/co-somatic colors require source-qualified genotype and explicit phasing.

Low-support ncORF sensitivity examples

Independent confirmation is required for both candidates. Reference residues remain neutral; only the focal differing RNA residue is orange.

MIR762HG | c16riboseqorf60 | complete RNA coding candidate

1 assembled fragments; 1 fragments each span the displayed coding interval; minimum base depth 1. SRR23456709; single read sensitivity.

REF	MQEDRKRCEYEWVSTTRMAKVKKTAPCGGEKVERL
RNA	MQEDRKRCEYEWVSTTTMAKVKKFRSV

RNA fragment support per codon (minimum of its three base depths)



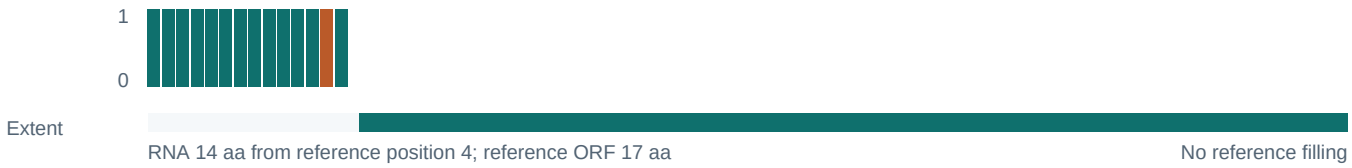
`Complete reference ORF: MQEDRKRCEYEWVSTTRMAKVKKTAPCGGEKVERLDLSCISGGV`
Single-read hypothesis. Nonfocal differences remain unclassified; a premature RNA stop or an apparent complete coding interval does not establish a translated product. Independent original-RNA confirmation is still needed.

PPIP5K2 | c5norep99 | partial RNA coding candidate

1 assembled fragments; 1 fragments each span the displayed coding interval; minimum base depth 1. SRR23456709; single read sensitivity.

REF	KNKRVFANSHHNIK
RNA	KNKRVFANSHHNMK

RNA fragment support per codon (minimum of its three base depths)



`Complete reference ORF: MISKNKRVFANSHHNIK`
Single-read hypothesis. Nonfocal differences remain unclassified; a premature RNA stop or an apparent complete coding interval does not establish a translated product. Independent original-RNA confirmation is still needed.

Focal differing residue Unclassified nonfocal difference RNA-covered interval. Reference residues stay neutral. Coverage outside the displayed RNA is unassessed.

Sources, processing and limits

All RNA libraries, DNA nominations, polished inputs and references across the report set are listed in SEQUENCING_SOURCES.md. Focused CDKN2A and EIF3K-CYP39A1 reports show highlighted reference alignments and original-RNA support separately from these atlas summaries.

Input	Pinned scope
Matched-normal small variants	SEQC2 v1.2.1 SNV + indel, GRCh38.d1.vd1; HCC1395BL normal
SV benchmark	Talsania et al. 2022 Table S4, HCC1395 vs HCC1395BL
Original RNA	PRJNA934933 / SRR23456709; 1,049,452 ONT direct RNA reads
Alignment	minimap2 2.26 -ax splice -uf -k14; NCBI GRCh38 no-alt
ncORF witness realignment	36 original ALT reads; GRCh38 full analysis set + decoy/HLA; same library

Reference model and interpretation

Small-variant coding effects use Varcode with Ensembl release 87 on GRCh38. The human orfdata 0.34.1 catalog uses Ensembl 101 GRCh38 reference ORFs (7,264 models, dataset 2026-10-11.1). The new ORFdata 0.39.1 replay validates nine complete catalogue ORF paths/proteins against Ensembl 87, then reconstructs their frames on mutation-supporting RNA through the shared library API. The ncORF chapter separates default and single-read results. Genomic transcript overlap is not RNA-compatible isoform support. Original-read allele counts and local protein contexts are separate observations.

SHA-256 input pins

SEQC2 SNV VCF gzip

`0c49f7e1519fc7860b773f79bdcf07453fe4def8303707f34542471d92494c9d`
SEQC2 indel VCF gzip

`db7dc8be2ec04bab830ff1411146c7d538c217ff684e874b85d98a3dbbda91cf`
Talsania Table S4 xlsx

`11e8d6393b83f0e0b7fdb0b13a57376663fc9662116d78c89a0d5fa530df096b`
SRR23456709 SRA archive

`d69db8e48bbb20665c3e03a1fa35742be818a0127994e015cb3a560afae39ac8`
GRCh38 no-alt FASTA (uncompressed)

`9cce8b926416dd96b152deea85188495b75f7ac8d634cc723a017067be8702b7`
ncORF realignment GRCh38 full analysis + decoy/HLA FASTA

`3b103f4742abfd54938fb0333e19ad067635c8eb86f1dbf0ce44b165c4292b50`
orfdata bundled GRCh38 catalog

`06bc0bf3f81c25a7356f71a38581ff5704c05dcb75752454bc1ca75075ab8c09`
Ensembl 87 GRCh38 protein FASTA gzip (self screen)

`d873c199d5dbf20869db3f2cbd6ac726259215d91c2782f9f1ddcad4bb117cf9`

What remains unresolved

ONT raw-read errors can create other-base calls and competing local contexts. The RNA and DNA libraries come from separate studies. A 500-base breakend window and split-alignment candidate are screening evidence only. Selected junctions were resolved with original reads, but complete isoform assignment and SV protein reconstruction remain unresolved.

Machine-readable per-call annotation, allele, protein, and SV tables and complete reconstruction commands are published with the Isovar HCC1395/COLO829 reports. These PDFs summarize selected public source call sets, not a complete catalogue of every cell-line mutation.

HCC1395 | EIF3K-CYP39A1 fusion

531-aa complete coding candidate - RNA reconstruction, not measured translation

149 exact Illumina junction fragments

9 ONT coding-span reads; 4 also have the exact junction

The reference-guided 1,596-nt coding RNA path is tiled by exact original reads, including 19 mapped Illumina reads with overlaps of at least 20 nt. The Illumina input is fusion-enriched. Counts do not measure whole-library prevalence.

IGV (GRCh37): chr19:39123318-39123318 chr6:46607405-46607405

Protein architecture and reference alignment

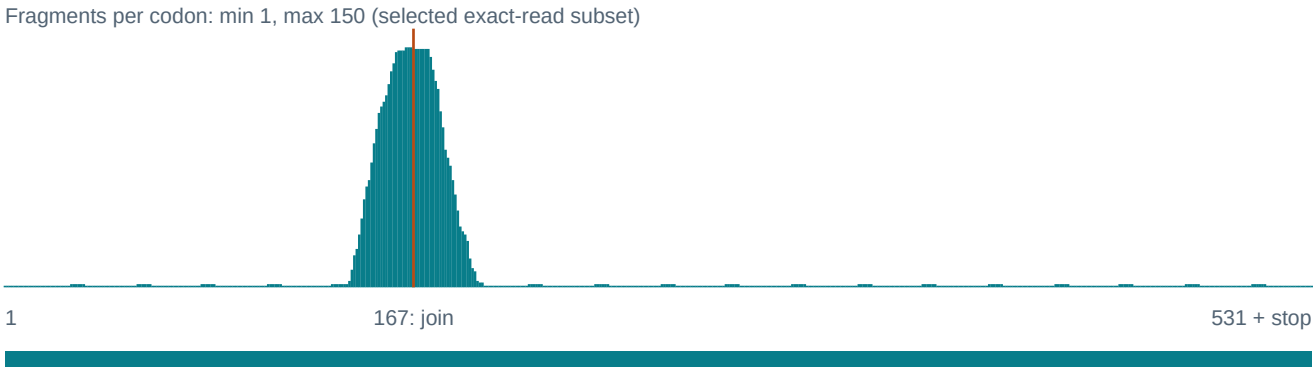
Fusion residues 1-166 match EIF3K residues 1-166. Residue 167 is the split junction codon: one EIF3K base plus two CYP39A1 bases. Its alanine matches CYP39A1 residue 105. Fusion residues 168-531 match CYP39A1 residues 106-469, followed by a stop.

Junction zoom: aligned by the pinned exon/CDS map

EIF3K 151-186	QHIDRWLLAEMLGDLSDSQLKVWMSKYGWSADESGQ
Fusion 151-186	QHIDRWLLAEMLGDLSASIPKNVFLALHEKLYIMLK
CYP39A1 89-124	KKVDFELAVQNIVYRTASIPKNVFLALHEKLYIMLK

The mutant highlight marks junction codon 167. The novel feature is the joined protein context; the CYP39A1-derived suffix is reference sequence, not a novel frameshift tail. Other reported fusion isoforms remain possible.

Exact original RNA support across the entire coding path



Assembly track: all 1,596 coding bases RNA-witnessed; 0 reference-filled bases.

Each bar counts distinct original fragment IDs with an exact read covering the whole codon. It is not raw BAM depth. The RNA library and ONT library remain separate. Six of 21 ONT candidate reads contain the exact 20+20-base junction; nine span coding start to stop in local collinear alignments. Four satisfy both checks.

HCC1395 | EIF3K-CYP39A1 fusion

Complete protein aligned to its two reference contributors

D = retained EIF3K reference; F = fusion candidate; A = retained CYP39A1 reference. Dashes mean the contributor is absent at that fusion position, not a deletion from the other gene. The fusion row distinguishes the CYP39A1 suffix and split junction codon. Coordinates at left refer to the fusion protein. The terminal * is the observed stop.

Fusion amino acids 1-60

```
D  MAMFEQMRANVGKLLKGIDRYNPENLATLERYVETQAKENAYDLEANLAVLKLYQFNPAF
F  MAMFEQMRANVGKLLKGIDRYNPENLATLERYVETQAKENAYDLEANLAVLKLYQFNPAF
A  ------------------------------------------------------------
```

Fusion amino acids 61-120

```
D  FQTTVTAQILLKALTNLPHTDFTLCKCMIDQAHQEERPIRQILYLGDLLETCHFQAFWQA
F  FQTTVTAQILLKALTNLPHTDFTLCKCMIDQAHQEERPIRQILYLGDLLETCHFQAFWQA
A  ------------------------------------------------------------
```

Fusion amino acids 121-180

```
D  LDENMDLLEGITGFEDSVRKFICHVVGITYQHIDRWLLAEMLGDLS--------------
F  LDENMDLLEGITGFEDSVRKFICHVVGITYQHIDRWLLAEMLGDLSASIPKNVFLALHEK
A  ---------------------------------------------ASIPKNVFLALHEK
```

Fusion amino acids 181-240

```
D  ------------------------------------------------------------
F  LYIMLKGKMGTVNLHQFTGQLTEELHEQLENLGTHGTMDLNNLVRHLLYPVTVNMLFNKS
A  LYIMLKGKMGTVNLHQFTGQLTEELHEQLENLGTHGTMDLNNLVRHLLYPVTVNMLFNKS
```

Fusion amino acids 241-300

```
D  ------------------------------------------------------------
F  LFSTNKKKIKEFHQYFQVYDEDFEYGSQLPECLLRNWSKSKKWFLELFEKNIPDIKACKS
A  LFSTNKKKIKEFHQYFQVYDEDFEYGSQLPECLLRNWSKSKKWFLELFEKNIPDIKACKS
```

Fusion amino acids 301-360

```
D  ------------------------------------------------------------
F  AKDNSMTLLQATLDIVETETSKENSPNYGLLLLWASLSNAVPVAFWTLAYVLSHPDIHKA
A  AKDNSMTLLQATLDIVETETSKENSPNYGLLLLWASLSNAVPVAFWTLAYVLSHPDIHKA
```

Fusion amino acids 361-420

```
D  ------------------------------------------------------------
F  IMEGISSVFGKAGKDKIKVSEDDLENLLLIKWCVLETIRLKAPGVITRKVVKPVEILNYI
A  IMEGISSVFGKAGKDKIKVSEDDLENLLLIKWCVLETIRLKAPGVITRKVVKPVEILNYI
```

Fusion amino acids 421-480

```
D  ------------------------------------------------------------
F  IPSGDLLMLSPFWLHRNPKYFPEPELFKPERWKKANLEKHSFLDCFMAFGSGKFQCPARW
A  IPSGDLLMLSPFWLHRNPKYFPEPELFKPERWKKANLEKHSFLDCFMAFGSGKFQCPARW
```

Fusion amino acids 481-531

```
D  -------------------------------------------------
F  FALLEVQMCIILILYKYDCSLLDPLPKQSYLHLVGVPQPEGQCRIEYKQRI
A  FALLEVQMCIILILYKYDCSLLDPLPKQSYLHLVGVPQPEGQCRIEYKQRI
```

Terminal stop: * | Annotated EIF3K start and CYP39A1 stop retained.

HCC1395 | EIF3K-CYP39A1 fusion

Full reference proteins and provenance of amino-acid changes

EIF3K / ENST00000248342

1-60	MAMFEQMRANVGKLLKGIDRYNPENLATLERYVETQAKENAYDLEANLAVLKLYQFNPAF
61-120	FQTTVTAQILLKALTNLPHTDFTLCKCMIDQAHQEERPIRQILYLGDLLETCHFQAFWQA
121-180	LDENMDLLEGITGFEDSVRKFICHVVGITYQHIDRWLLAEMLGDLSDSQLKVWMSKYGWS
181-218	ADESGQIFICSQEESIKPKNIVEKIDFDSVSSIMASSQ

CYP39A1 / ENST00000275016

1-60	MELISPTVIIILGCLALFLLLQRKNLRRPPCIKGWIPWIGVGFEFGKAPLEFIEKARIKY
61-120	GPIFTVFAMGNRMTFVTEEEGINVFLKSKKVDFELAVQNIVYRTASIPKNVFLALHEKLY
121-180	IMLKGKMGTVNLHQFTGQLTEELHEQLENLGTHGTMDLNNLVRHLLYPVTVNMLFNKSLF
181-240	STNKKKIKEFHQYFQVYDEDFEYGSQLPECLLRNWSKSKKWFLELFEKNIPDIKACKSAK
241-300	DNSMTLLQATLDIVETETSKENSPNYGLLLLWASLSNAVPVAFWTLAYVLSHPDIHKAIM
301-360	EGISSVFGKAGKDKIKVSEDDLENLLLIKWCVLETIRLKAPGVITRKVVKPVEILNYIIP
361-420	SGDLLMLSPFWLHRNPKYFPEPELFKPERWKKANLEKHSFLDCFMAFGSGKFQCPARWFA
421-469	LLEVQMCIILILYKYDCSLLDPLPKQSYLHLVGVPQPEGQCRIEYKQRI

Highlighted reference residues contribute to the fusion. Dimmed residues are not retained. The fusion codon at residue 167 contains bases from both partners and encodes CYP39A1 alanine 105; it is not counted as a complete EIF3K-derived codon.

Variant provenance

- Focal change
- Germline
- Phased somatic

No additional non-reference residues occur in the displayed RNA sequence relative to its indicated reference contributor(s). No germline or other somatic allele is assigned without genotype and phasing evidence. Missing sequence stays unobserved.

There is no reference-filled region in the determined ORF. The gene junction at coding offset 499 is an RNA-supported splice/fusion boundary, not an assembly-to-reference stitching point. Support falls outside the enriched junction region, as shown by the codon track; no junction count is assigned to every residue.

HCC1395 | EIF3K-CYP39A1 fusion

All sequencing and reference inputs used in this focused report

Illumina tumor RNA: public INTEGRATE test-data.tar.gz

Original unmapped.bam (185 fusion-enriched records) and accepted_hits.bam (19 mapped records retained for coding tiling). Archive SHA-256: 9a37130e768e424935008f1d0a67050f6e11a242a263ed2f20ba071d94b767cb. The archive is the source identifier; no SRA run accession is asserted for this subset.

Independent ONT direct RNA: SRR23456709 / PRJNA934933

BioSample SAMN33292477; 1,049,452 reads in the source run. The report checks 21 original candidates and 42 local alignment records. This is a local candidate screen, not a genome-wide uniqueness test.

Tumor and matched-normal DNA: INTEGRATE example archive

All original alignments in two GRCh37 flanking intronic windows were retained. 18 tumor cross-gene fragments versus 0 normal. This supports a rearrangement, without resolving its exact DNA breakpoint.

Exon/cDNA and coding references

Pinned GRCh37-lite archive models: EIF3K ENST00000248342 and CYP39A1 ENST00000275016. Ensembl 87 CDS offsets transfer only after exact whole-transcript sequence identity. ONT placements use GRCh38 and separately pinned Ensembl 87 exon coordinates; genomic coordinates from the two assemblies are never mixed.

Derived files, checksums and broader report source index

The adjacent HCC1395_EIF3K_CYP39A1_fusion.json pins all eight BAM/FASTA/model inputs and contains the complete protein, reference contributors and per-codon support. SEQUENCING_SOURCES.md lists every library, DNA source and reference used in the Sid/HCC1395/COLO829 report set.

The first page highlights strong junction support. It does not assign 149 fragments to every amino acid or claim that 149 reads span the entire protein. Complete coding sequence support is the separately displayed overlap tiling, corroborated by the source-separated ONT checks.